CRYPTOCURRENCY ANALYTICS REPORT
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1. OVERVIEW

This report documents the end-to-end process of building a cryptocurrency analytics dashboard in Power BI, including data preparation, modeling, DAX calculations, and visualization techniques used to analyze price trends, volatility, trading volume, and sentiment (Fear & Greed Index).

Field Descriptions (First Dataset: consolidated_coin_data)

1. Currency

- This column identifies which cryptocurrency the row belongs to.
- Example values: *Bitcoin (BTC), Ethereum (ETH), Tezos (XTZ)*
- Used for filtering, grouping and comparing different coins.

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2. Date

- The trading date of the record.
- Each row represents one day of trading activity for that coin.
- Used for time-series analysis, charts, monthly/weekly summaries, and trends.

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3. Open

- The price at which the cryptocurrency **started trading on that day**.
- Helps identify if the price moved up or down compared to the close of the previous day.

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4. High

- The **highest price** reached during that day.
- Used for calculating volatility, risk, and intraday movement.

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5. Low

- The **lowest price** of the day.
- Together with High, it measures how much the price fluctuated.

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6. Close

- The **final price** at the end of the trading day.
- This is the most important metric for analysis because:
    - Returns (daily, monthly) use the closing price.
    - Trend lines and moving averages use Close.
    - Volatility and comparison visuals rely on it.

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7. Volume

- The total amount of the cryptocurrency traded during the day.
- Shows how active the market was:
    - High volume = strong interest, possible trend confirmation
    - Low volume = weak interest, uncertain trend

Used in correlation analysis (Volume → Price movements).

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8. MarketCap

- Market Capitalization = **Total value of the coin** in circulation.
- Formula generally: *Price × Circulating Supply*.
- Indicates:
    - Size of the coin
    - Investor confidence
    - Market dominance

Used to compare large, mid, and small-cap coins.

2. BUSINESS MODELING

The goal of the model is to provide traders, analysts, and investors with a unified space to understand crypto behavior across time. The business questions involve:
- How does the closing price of each cryptocurrency change over time?
- What is the volatility level for each coin?
- How do trading volume and price correlate?
- What months generate positive or negative returns?
- How does market sentiment (Fear & Greed Index) relate to price movements?

3. DATA PREPARATION STEPS

1. Import datasets: consolidated_coin_data and FearGreed.
2. Clean data:
   - Remove nulls, Dups & Negatives.
   - Set proper data types.
   - Fix date formatting.
   - Normalize column names.
3. Create DateTable including YearMonth, YearMonthDay, and MonthIndex.
4. Create relationships:
   - consolidated_coin_data[Date] → DateTable[Date]
   - FearGreed[LocalDate] → DateTable[Date]

4. KEY CALCULATED TABLES & COLUMNS

- YearMonth_AvgClose: summarized monthly average closing price per cryptocurrency.
- PrevMonthAvgClose: previous month's average value for return calculations.
- MonthlyReturn%: percentage return relative to previous month.
- Volatility column: High - Low.
- Converted local timestamp for FearGreed sentiment data.

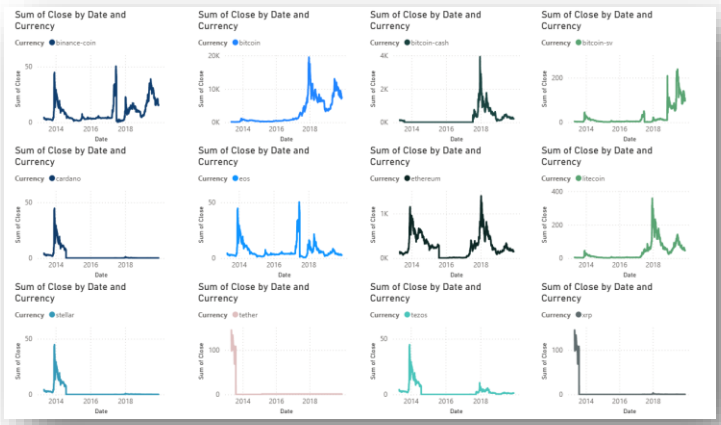
5. DAX MEASURES USED

Examples:
- Volatility = AVERAGE(consolidated_coin_data[High] - consolidated_coin_data[Low])
- HighestPrice = MAX(consolidated_coin_data[High])
- LowestPrice = MIN(consolidated_coin_data[Low])
- MonthlyReturn% = DIVIDE(AvgClose - PrevMonthAvgClose, PrevMonthAvgClose)

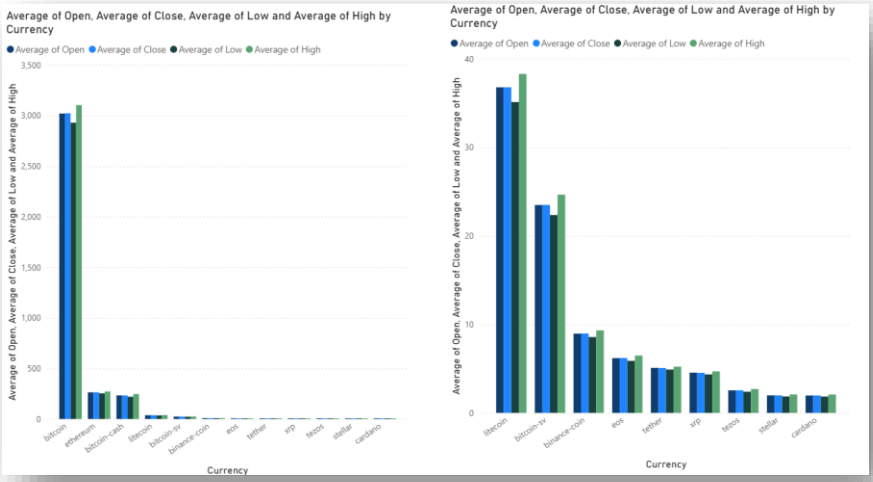
6. VISUALIZATION LAYER

Created Power BI visuals:
- Line chart: Date vs Close
- Column chart: Average price comparison

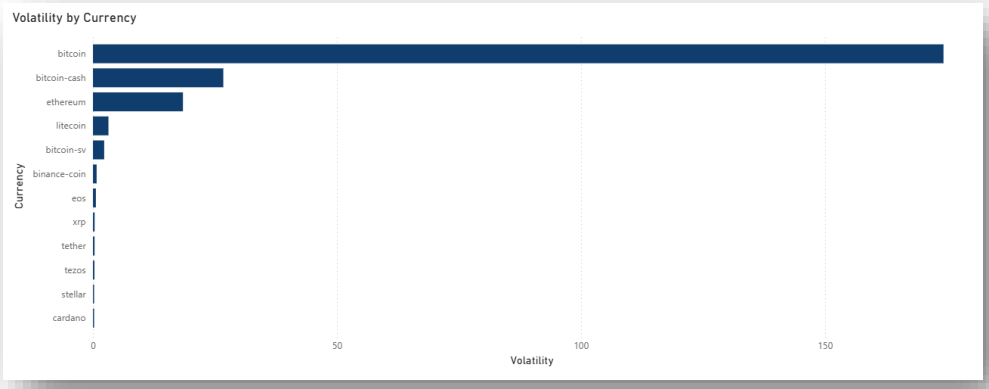
- KPI cards: Highest/Lowest prices
- Scatter chart: Volume vs Close
- Bar chart: Price Volatility
- Slicers: Currency, Year, Month
- Conditional formatting for highlighting returns, volatility, and sentiment.



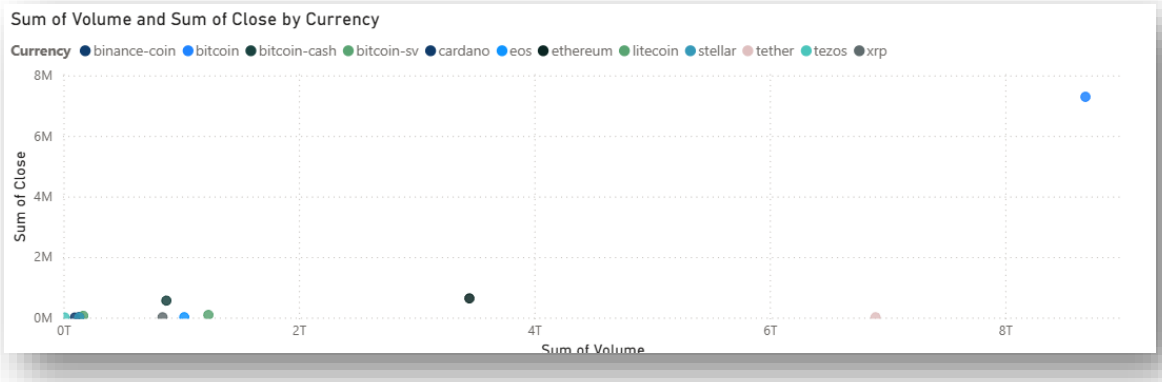
Sum of Close in Separated coins



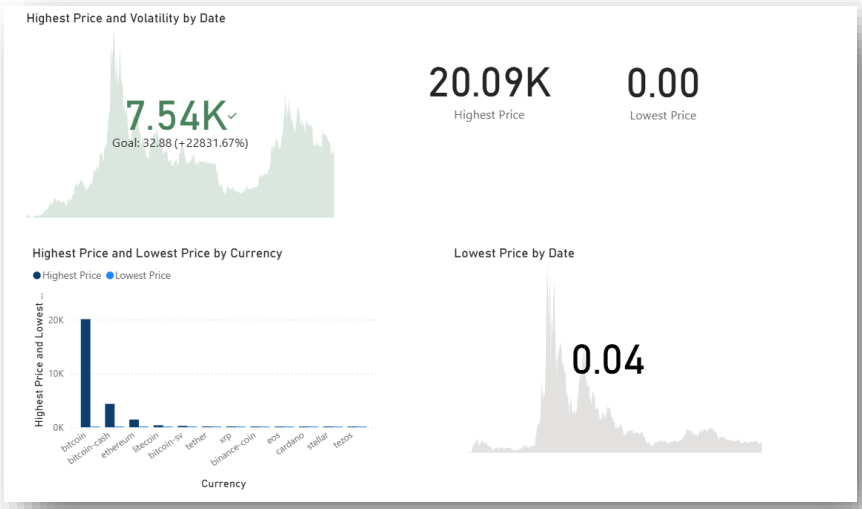
Average of Open, close, Hight, Low price in every coin that shows Bitcoin is top-1 with long distance from another coins.



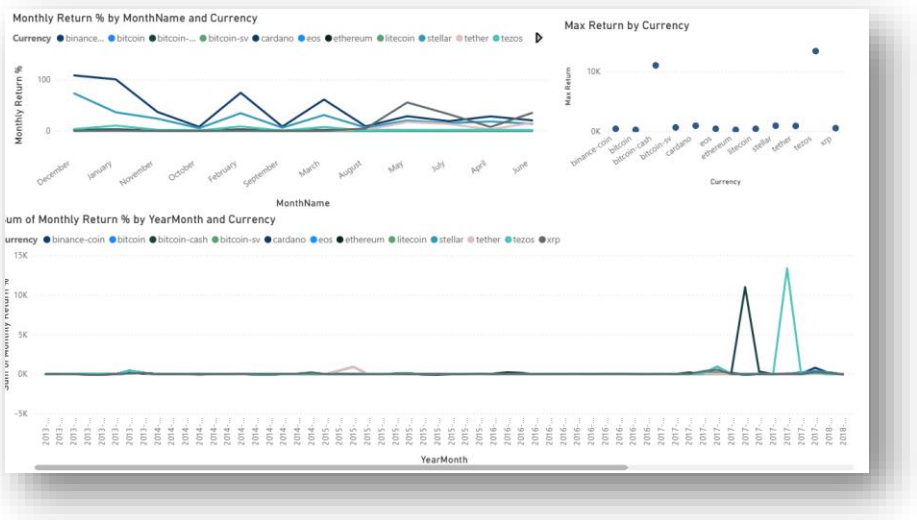
Volatility by Currency That shows Despite experiencing heavy selling with large volumes, Bitcoin has also shown high volatility.



Naturally, an increase in volume usually leads to a rise in the closing price, but the important factor is the intensity of this increase. Since a cryptocurrency like Bitcoin has gained considerable popularity, with rising volume, we observe that the closing price also increases exponentially. However, for certain cryptocurrencies like Tether, which have more controlled transaction volumes, we see that with increasing volume, the price converges to a fixed value. This indicates stability and lower risk.



For see the market status



These charts show that the monthly increase of cryptocurrencies like Bitcoin was not as high as expected. However, some other coins experienced monthly percentage increases of over 1000% before subsequently dropping.

7. DASHBOARD

PAGE 1 — Overview

1. KPI Cards

- **Sum of MarketCap (208T)**
  Shows the **total market capitalization** of all selected cryptocurrencies.
    - Tells you **overall market size**.
    - Useful for comparing different time periods or currency filters.
- **Average of Volume (813.31M)**
  Shows the **average trading volume**.
    - High volume = strong market activity
    - Low volume = weak interest, low liquidity
- **Max Monthly Return % (13.37K)**
  The **highest single-month return** among the selected currencies.
    - Helps identify **best performing periods**.

- **Min Monthly Return % (-99.92)**
  Shows the **worst month** — indicates **risk** and **loss potential**.

PAGE 2 — Price Trends

2. Line Chart: Average Close by Month & Currency

- Shows the **price evolution over years (2013–2019)**.
- Helps detect:
  - Bull markets
  - Bear markets
  - Growth patterns
  - Seasonality or cyclic behavior

Standard Deviation of Close (1.37K)

- Measures **volatility** in prices.
- High SD → unstable, risky currency
- Low SD → stable, lower risk

PAGE 3 — Comparison (Part 1)

3. Line Charts

- **Sum of Volume by Date**
  Shows **liquidity intensity** of each currency.
- **Sum of MarketCap by Date**
  Shows **long-term strength**, adoption, and decline phases.

What this page tells us

These visuals allow you to:

- Compare **popularity** and **liquidity** between coins
- Track **rise and fall** of market value
- Identify which currency is **growing**, **declining**, or **stable**

PAGE 4 — Comparison (Part 2)

4. Bar Charts

- **Average of Open/Close/Low/High**
  Compares price levels across currencies.
- **Average of Volume & MarketCap**
  Shows:
    - Which coin is traded the most
    - Which has the highest valuation
- **Volatility by Currency**
  A volatility comparison bar chart.
    - Important for risk analysis
    - High volatility = higher risk & reward
    - Low volatility = stability, lower returns

PAGE 5 — Insights (Part 1)

5. Bar + Line Combo

- **Volume vs Close price trend together**
    - Shows whether volume spikes **drive price changes**

Scatter Plot

- **Average Close vs Average Volume**
    - Shows relationship between **liquidity and value**
    - Helps identify:
        - Undervalued coins (high value, low volume)
        - Overhyped coins (high volume, low value)

PAGE 6 — Insights (Part 2)

6. Line Chart + Pie Chart

- **Sum of Avg Close**
  Shows long-term price trends for a chosen coin.
- **Pie Chart of Avg Monthly Close**
  Shows how much each currency contributes to overall market performance.
- **Monthly Returns Chart**
  - Shows profitability per month
  - Helps spot **most profitable times of the year**

PAGE 7 — Insights (Part 3)

7. Single Currency Trend (Close, MarketCap, Volume)

- You can switch metrics (Close / MarketCap / Volume).
- Shows micro-level:
  - Price cycles
  - MarketCap rise/fall
  - Trading activity

PAGE 8 — Fear & Greed

8. Scatter Plot of Fear & Greed (FG Index)

- Plot of **FG Volatility vs FG Close**
- Shows emotional market behavior.
  - "Fear" → lower prices, high volatility
  - "Greed" → higher prices, low volatility

Bar Chart — FG Scores by Month/Day

- Shows market psychology over time.
- Helps identify:
  - Panic phases

    - Greed-driven bull runs

Donut chart — Value Classification

- Shows distribution of:
    - Extreme Fear
    - Fear
    - Neutral
    - Greed
    - Extreme Greed

SUMMARY

Your dashboard provides a **complete market intelligence suite**:

Page	**Key Purpose**
Overview	High-level market health
Price Trends	Trend + volatility
Comparison	Compare currencies on volume, price, marketcap
Insights	Analytical relationships, anomalies, seasonal patterns
Fear & Greed	Psychological sentiment, predicting reversals

7. SENTIMENT INTEGRATION (FEAR & GREED INDEX)

Converted UNIX timestamps to local datetime, aligned sentiment date with trading date, and merged relevant sentiment fields including 'value' and 'value_classification'. This allows correlation analysis between sentiment and price movement.

8. FINAL INSIGHTS

- Volatility patterns help identify risky periods.
- Monthly returns reveal consistent growth/decline trends.
- Sentiment scores often align with market turning points.
- Volume-price correlation identifies accumulation/distribution phases.

In general, if you are looking to invest for the long term, Bitcoin is definitely the best choice. If you need stability, Tether is suitable, and if you want profit with lower risk and less volatility than Bitcoin, I recommend Ethereum.